SIMATS ENGINEERING

ASSESSMENT TOOL 3:

MCQ Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1~~9~~6

COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications.* (BL3)

Dave's Taxonomy: Precision (Level 3) **Total Marks: 30**

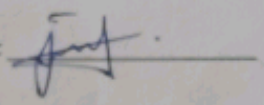
Question Count: 15

Duration : 60 Minutes

Code of Conduct

I Jannathul Firdous.N (192573025) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

MCQ Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Understanding	30%	Answers reflect strong and accurate concept mastery	Mostly accurate understanding	Partial understanding	Weak understanding
Accuracy of Responses	30%	Nearly all responses are correct	Most responses are correct	Some responses are correct	Few responses are correct
Application Awareness	15%	Shows strong recognition of practical cloud use cases	Reasonable recognition	Limited recognition	Weak recognition
Time Management	10%	Completes assessment efficiently	Completes on time	Needs extra time	Unable to finish well
Question Interpretation	15%	Interprets all items correctly	Misreads very few items	Misreads several items	Frequently misinterprets questions

Total Marks Awarded:

MCQ. 103 AT3

A company's cloud storage system requires 5TB of data replication across 3 regions what is the total storage required? c) 15TB

2. A cloud storage provider offers a 99.99% uptime SLA. How much downtime can be expected per month. a) ~ 4 min
a) ~4 minutes b) ~ 40 min c) ~4 hrs d) ~24 hrs.

3. 60,000 requests/sec among 10 servers. what is the average number of requests per server/sec.
b) 6000

4. 1 Gbps bandwidth and needs to transfer 500 GB of data how long it will take. c) 1.1 hrs.
a) 1 hr b) 10 hr c) 1.1 hrs d) 5.5 hrs

5. 99.95% availability rate. How much downtime does it experience per year? a) ~4 hrs.
a) ~4 hrs b) ~8 hrs c) ~20 hrs d) ~40 hrs.

6. A cloud provide offer at $0.05 per GB of storage If a company stores 10 TB. What is the monthly cost? b) $500
a) $50 b) $500 c) $1000 d) $2000

7. 10 Gbps throughput 2% packet loss. what is effective throughput. a) 9.8 Gbps

a) 9.8 Gbps b) 9.6 Gbps c) 8.0 Gbps d) 7.5 Gbps

8. A user upload 250 MB of data per day to the cloud how much data will be uploaded in 1 year (365 days) a) 91.25 GB

a) 91.25 GB b) 65 GB c) 100 GB d) 120 GB

9. 5 Mbps bandwidth per user. If 100 users are streaming simultaneously. what is the total required bandwidth. a) 500 Mbps

a) 500 Mbps b) 250 Mbps c) 200 Mbps d) 1000 Mbps.

10. A cloud provider guarantees 99.95% uptime per year. how much downtime does this allow in minutes per year? a) 262 min

a) 262 min b) 365 min c) 525 min d) 150 min.

11. Data center has 200 physical server If each server has 40 VM. what is the total no of VM.? b) 8000

a) 4000 b) 8000 c) 6000 d) 10,000

12. A physical server has 128 GB of RAM and hosts 10 VM. each require 12 GB how much memory is overcommitted? c) - 8 GB

a) 4GB b) 6GB c) - 8GB d) 10 GB

13. A virtual machine requires 1 vCPU. If a server has a total of 64 cores and an overcommitment ratio of 2:1 how many VMs can be hosted? b) 32

a) 16 b) 32 c) 64 d) 128

14. A company migrates 50 on permise servers to the cloud reducing power consumption by 60%. If original power is 100 kW what is new power consumption? c) 60 kW

a) 40 kW b) ~~50~~ kW c) 60 kW d) 70 kW

15. 100 physical server each with 64 vCPUs. overcommitment ratio 4:1 · how many total vCPUs can be allocated?

a) 6400 b) 12,800 c) 25,600 d) 50,000 c) 25,600

16. Private and public cloud resources?

a) hybrid cloud.

17. Sensitive data remains private even if intercepted
a) Encryption.

18. Hypervisor hosts 50 VMs and each VM uses 20 GB bandwidth per day calculate total bandwidth consumption per week. b) 7000

19. A cloud system runs 70% average CPU utilization. If has 32 cores at 2.5 GHz each total utilized processing power? a) 56 GHz
a) 56 GHz b) 56 GHz c) 64 GHz d) 70 GHz.

20. If network latency in a virtualized system is 5ms and processing time is 10ms calculate total round trip delay for 100 requests
a) 1.5 sec
a) 1.5 sec b) 2 sec c) 1 sec d) 3 sec.